

Palm Oil Retention & Melting System

Submerged Hydronic Coil Design — Technical Proposal II

Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

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This document presents thermal sizing, heat loss analysis, NTU- ϵ performance evaluation, process control architecture, and production capacity validation for a 3×10 kL palm oil storage and 1×5 kL replenishment tank system, integrated with a passive solar thermal loop and auxiliary electric immersion heating.

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Executive Summary

This document presents the thermal design and sizing of a submerged hydronic coil system for the retention, melting, and trim-heating of palm oil at a Kathmandu processing facility, developed as part of an internal engineering study at Mechatronics Engineering Pvt. Ltd. (MEPL). The system comprises three 10 kL retention tanks ($T_{10,A-C}$) that receive and melt solid palm oil, and a single 5 kL insulated replenishment tank (T_5) that trim-heats the molten oil to the 60 °C process target. Heat is supplied primarily by a passive solar thermal loop, backed by a 3×12 kW electric immersion auxiliary battery for off-solar hours. The work presented here — covering vessel heat loss, coil and water-side hydraulics, two-phase U -value determination, NTU- ϵ performance, energy and melt-time budgeting, and process control architecture — was independently researched, modelled, and authored by Sambridda Ranabhat.

Headline Results

Metric	Result
Kathmandu annual solar reliability	75.9%
Single-tank capacity (primary, 75 °C)	(Depends on Pipe configuration)
Single-tank capacity (worst case, 70 °C)	(Depends on Pipe configuration)
Safety margin vs. 2 t/day target	$2.42\times - 3.63\times$
Effective capacity, 3 tanks staggered	≥ 20 t/day (10 $\times$ margin)
Tank heat loss, melt phase (uninsulated)	≈ 1.07 kW
Overall U — solid / liquid phase	(SEE Table 7)
T_5 outlet temperature vs. target	(Varies between 55 °C and 40 °C)

Key Findings

- **The 2 t/day target is met with substantial margin** even under the worst-case 70 °C thermal driver scenario, using a single uninsulated T_{10} tank. With all three tanks staggered, the system delivers a 10 $\times$ margin over target.
- **The solid-phase heat transfer coefficient is the system bottleneck**, governing the ≈ 28 –33 hr melt time per tank. Mechanical agitation or a higher driver temperature are identified as the primary levers if future throughput targets increase.
- **T_5 is thermally over-specified** for its sensible-heating duty ($NTU \approx 20$), so a duty-cycle loop is required to hold the Dynamic Upper Thermal Boundary of $T_f(x, y, m_p) > 55$ °C based on dynamic set point rather than static set point preventing overshooting above 60 °C.
- **The control system is fully demand-driven:** T_5 is the process master, T_{10} tanks are ranked continuously by a volume/temperature-weighted scoring system, and oil distribution to downstream lines is priority-queued and HMI-configurable.
- **Insulating T_{10}** is recommended as a low-cost efficiency gain, reducing hold-phase losses by $\approx 85\%$.

The remainder of this document develops each of these results from first principles, beginning with the Kathmandu climate baseline that underpins every thermal calculation that follows.

1 System Overview

1.1 Scope and Purpose

This proposal sizes the hydronic heating coil system for a palm oil storage and batch processing facility in Kathmandu, Nepal. The system must:

1. Receive solid palm oil (stored at ambient $\approx 18.1^\circ\text{C}$) into three 10 kL retention tanks ($T_{10,A}$, $T_{10,B}$, $T_{10,C}$), melt it, and hold it in a liquid state.
2. Transfer liquid palm oil via a high-pressure pump (HPP) to a single 5 kL insulated replenishment tank (T_5), where it is trim-heated to the process target of 60°C .
3. Achieve a daily throughput target of ≥ 2 tonnes of processed liquid palm oil.
4. Derive primary heat from a passive solar thermal loop ($5\times$ evacuated-tube collectors, 2500 L reservoir) with an auxiliary thermal battery of $3 \times 12\text{ kW}$ SS 316L immersion elements for off-solar hours.

1.2 Vessel Summary

Table 1: Vessel inventory and key parameters.

Tag	Role	Volume	Dia. (m)	Cyl. Ht. (m)	Bottom	Insulated
$T_{10} (\times 3)$	Long-term storage / melting	10 kL	2.32	2.20	Flat	No
T_5	Replenishment / trim-heat	5 kL	1.80	2.00	Conical	Yes

1.3 Thermal Driver

The primary thermal driver is the solar loop reservoir, which delivers hot water at a weighted annual average of $\approx 76^\circ\text{C}$ – 78°C based on the Kathmandu seasonal irradiance analysis. For this sizing, the following design temperatures are used:

- **Primary case:** $T_{\text{hot}} = 75^\circ\text{C}$
- **Worst case:** $T_{\text{hot}} = 70^\circ\text{C}$

The paper calculates for the delivery medium to be hot water supplied via 1 inch Schedule 40 CPVC pipe from the thermal battery to the tank coil headers. A single supply pipe size is used throughout the system; the coil (water-side) pipe bore is treated as the principal sizing variable and is evaluated across the standard 1 inch–2 inch NPS range.

2 Kathmandu Climate Baseline and Solar Thermal Achievability

2.1 Weighted Average of Usable Solar Days

To determine the long-term reliability of the solar utility system in Kathmandu, seasonal irradiance data is aggregated into a single annual usability metric. The weighted average is computed by dividing the total usable solar days by the total days in the year:

$$\bar{U} = \frac{\sum_i (\text{Days}_i \times \text{Usable}\%_i)}{\sum_i \text{Days}_i} = \frac{\text{Total Usable Days}}{\text{Total Days}} \quad (1)$$

Substituting the seasonal station metrics:

Total Usable Days = 73 (Winter) + 82 (Spring) + 67 (Monsoon) + 55 (Autumn) = 277 days

Total Days = 90 + 92 + 122 + 61 = 365 days

$$\bar{U} = \frac{277}{365} \approx 0.7589 \implies \mathbf{75.9\% \text{ annual solar reliability}} \quad (2)$$

2.2 Thermal Achievability of the 75 °C Loop Target

To confirm that the utility loop can reach and sustain 75 °C without solar input, a dead-start worst case is evaluated. The auxiliary system consists of 3 × 12 kW SS 316L immersion elements ($P_{\text{aux}} = 36 \text{ kW}$ total) acting through an SSR-modulated controller on a 300 L insulated active buffer volume. The initial temperature is taken as ambient: $T_{\text{init}} = 24 \text{ °C}$.

Total Thermal Energy Requirement

$$Q = m C_p \Delta T = 300 \text{ kg} \times 4184 \text{ J kg}^{-1} \text{K}^{-1} \times (75 - 24) \text{ K} = 64.0 \text{ MJ} \quad (3)$$

Time-to-Temperature Response

$$t = \frac{Q}{P_{\text{aux}}} = \frac{64,015,200 \text{ J}}{36,000 \text{ W}} \approx 1778 \text{ s} \approx 29.6 \text{ min} \quad (4)$$

The auxiliary electric system can drive the entire active buffer from ambient to 75 °C in under 30 minutes from a cold start, confirming that maintaining a steady-state 75 °C process loop is operationally stable and achievable even during zero-solar periods.

2.3 Weighted Average Ambient Temperature — Kathmandu

The ambient design temperature used throughout this proposal ($T_\infty = 18.1^\circ\text{C}$) is derived from the Kathmandu historical monthly average, weighted by the number of days in each month.

$$\bar{T}_\infty = \frac{\sum_{i=1}^{12} (x_i \cdot w_i)}{\sum_{i=1}^{12} w_i} \quad (5)$$

where x_i is the monthly average temperature and w_i the number of days in month i .

Table 2: Kathmandu monthly temperature data and weighted contribution.

Month	Avg Temp x_i ($^\circ\text{C}$)	Days w_i	Weighted $x_i w_i$
January	9.4	31	291.4
February	12.2	28	341.6
March	16.1	31	499.1
April	20.0	30	600.0
May	22.2	31	688.2
June	23.9	30	717.0
July	23.3	31	722.3
August	23.3	31	722.3
September	22.2	30	666.0
October	19.4	31	601.4
November	14.4	30	432.0
December	10.6	31	328.6
Total		365	6609.9

$$\bar{T}_\infty = \frac{6609.9}{365} = 18.109 \dots \approx 18.1^\circ\text{C} \quad (6)$$

This value is used as T_∞ in all tank heat loss and initial oil temperature calculations throughout this document.

3 Vessel Geometry

3.1 T₁₀ (10 kL Retention Tank)

Tank T₁₀ is a mild-steel vessel with a cylindrical body and a flat bottom. The external surface area governs the heat loss calculation.

$$r_{10} = \frac{D_{10}}{2} = \frac{2.32}{2} = 1.16 \text{ m} \quad (7)$$

$$A_{\text{cyl}} = \pi D_{10} H_{10} = \pi \times 2.32 \times 2.20 = 16.035 \text{ m}^2 \quad (8)$$

$$A_{\text{base}} = \pi r_{10}^2 = \pi \times 1.16^2 = 4.229 \text{ m}^2 \quad (9)$$

$$A_{\text{vessel},10} = A_{\text{cyl}} + A_{\text{base}} = 16.035 + 4.229 = 20.264 \text{ m}^2 \quad (10)$$

3.2 T₅ (5 kL Replenishment Tank)

Tank T₅ is a mild-steel vessel with a cylindrical body and a conical bottom (half-angle 30°).

$$r_5 = 0.90 \text{ m} \quad (11)$$

$$h_{\text{cone},5} = r_5 \tan(30) = 0.90 \times 0.5774 = 0.520 \text{ m} \quad (12)$$

$$\ell_{\text{slant},5} = \sqrt{r_5^2 + h_{\text{cone},5}^2} = \sqrt{0.90^2 + 0.520^2} = 1.040 \text{ m} \quad (13)$$

$$A_{\text{cyl},5} = \pi D_5 H_5 = \pi \times 1.80 \times 2.00 = 11.310 \text{ m}^2 \quad (14)$$

$$A_{\text{cone},5} = \pi r_5 \ell_{\text{slant},5} = \pi \times 0.90 \times 1.040 = 2.940 \text{ m}^2 \quad (15)$$

$$A_{\text{vessel},5} = A_{\text{cyl},5} + A_{\text{cone},5} = 11.310 + 2.940 = 14.250 \text{ m}^2 \quad (16)$$

4 Tank Heat Loss Analysis

4.1 Methodology

T₁₀ is uninsulated mild steel ($t_{\text{wall}} = 6 \text{ mm}$, $k_{\text{steel}} = 50 \text{ W m}^{-1} \text{ K}^{-1}$). Heat is lost by: (1) inner-surface convection from oil to wall, (2) conduction through the steel, (3) natural convection from the outer wall to ambient air. Radiation is excluded per design brief. Two loss conditions are evaluated: the *melting phase* ($T_{\text{oil}} = 36^\circ \text{C}$), which gives the minimum loss rate and is used to net out the coil's useful melting output (Section ??); and the *maximum-temperature* condition ($T_{\text{oil}} = 75^\circ \text{C}$, the PLC high-limit set point — see Table 14), which gives the conservative upper-bound loss rate used to define the coil sizing target Q_{target} in Section 8.

Ambient temperature: $T_\infty = 18.1^\circ \text{C}$ (Kathmandu weighted annual average).

4.2 Conduction Resistance

$$R_{\text{cond}} = \frac{t_{\text{wall}}}{k_{\text{steel}}} = \frac{0.006}{50} = 1.20 \times 10^{-4} \text{ m}^2 \text{ K W}^{-1} \quad (17)$$

4.3 External Natural Convection — Cylindrical Wall (Melting Phase, 36°C)

Air properties evaluated at film temperature $T_f = (36 + 18.1)/2 = 27.1^\circ \text{C}$: $\mu_{\text{air}} = 1.849 \times 10^{-5} \text{ Pa s}$, $k_{\text{air}} = 0.02624 \text{ W m}^{-1} \text{ K}^{-1}$, $\text{Pr}_{\text{air}} = 0.7296$, $\nu_{\text{air}} = 1.571 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, $\beta_{\text{air}} = 1/T_f(\text{K}) = 3.322 \times 10^{-3} \text{ K}^{-1}$.

Rayleigh number (characteristic length $L = H_{10} = 2.20 \text{ m}$):

$$Ra_L = \frac{g \beta_{\text{air}} \Delta T_{\text{vessel}} L^3}{\nu_{\text{air}}^2} \Pr_{\text{air}} = \frac{9.81 \times 3.322 \times 10^{-3} \times 17.9 \times 2.20^3}{(1.571 \times 10^{-5})^2} \times 0.7296 = 1.841 \times 10^{10} \quad (18)$$

Churchill–Chu correlation for a vertical surface:

$$\overline{Nu}_L = \left[0.825 + \frac{0.387 Ra_L^{1/6}}{[1 + (0.492/\Pr)^{9/16}]^{8/27}} \right]^2 = \left[0.825 + \frac{0.387 \times (1.841 \times 10^{10})^{1/6}}{1.102} \right]^2 = 307.3 \quad (19)$$

$$h_{\text{ext,cyl}} = \frac{307.3 \times 0.02624}{2.20} = 3.665 \text{ W m}^{-2} \text{ K}^{-1} \quad (20)$$

4.4 External Natural Convection — Flat Base (Melting Phase, 36°C)

The flat circular base is warmer than ambient and faces downward (hot surface facing down), so buoyancy suppresses convection. The Morgan–McAdams correlation for a heated horizontal plate facing downward applies, with characteristic length $L_c = A_{\text{base}}/P_{\text{base}} = r_{10}/2 = 0.58 \text{ m}$:

$$Ra_{L_c} = \frac{g \beta_{\text{air}} \Delta T_{\text{vessel}} L_c^3}{\nu_{\text{air}}^2} \Pr_{\text{air}} = \frac{9.81 \times 3.322 \times 10^{-3} \times 17.9 \times 0.58^3}{(1.571 \times 10^{-5})^2} \times 0.7296 = 3.374 \times 10^8 \quad (21)$$

$$\overline{Nu}_{L_c} = 0.27 Ra_{L_c}^{0.25} = 0.27 \times (3.374 \times 10^8)^{0.25} = 36.59 \Rightarrow h_{\text{ext,base}} = \frac{36.59 \times 0.02624}{0.58} = 1.656 \text{ W m}^{-2} \text{ K}^{-1} \quad (22)$$

4.5 Overall Heat Loss Coefficients and $\dot{Q}_{\text{loss}}$ — Melting Phase

Inner surface convection during solid/melting phase: $h_{\text{in,vessel}} = 35 \text{ W m}^{-2} \text{ K}^{-1}$ (conduction-dominated, stagnant melt sleeve — see Section 7).

$$U_{\text{ves,cyl}} = \left(\frac{1}{35} + 1.20 \times 10^{-4} + \frac{1}{3.665} \right)^{-1} = 3.317 \text{ W m}^{-2} \text{ K}^{-1} \quad (23)$$

$$U_{\text{ves,base}} = \left(\frac{1}{35} + 1.20 \times 10^{-4} + \frac{1}{1.656} \right)^{-1} = 1.581 \text{ W m}^{-2} \text{ K}^{-1} \quad (24)$$

$$\dot{Q}_{\text{loss,cyl}} = U_{\text{ves,cyl}} \times A_{\text{cyl}} \times \Delta T = 3.317 \times 16.035 \times 17.9 = 951.9 \text{ W} \quad (25)$$

$$\dot{Q}_{\text{loss,base}} = U_{\text{ves,base}} \times A_{\text{base}} \times \Delta T = 1.581 \times 4.229 \times 17.9 = 119.7 \text{ W} \quad (26)$$

$$\dot{Q}_{\text{loss,melt}} = 951.9 + 119.7 = 1071.6 \text{ W} \approx 1.07 \text{ kW} \quad (27)$$

4.6 External Natural Convection and $\dot{Q}_{\text{loss}}$ — Maximum-Temperature Condition (75°C)

To establish a conservative coil-sizing target (Section 8), the same methodology is repeated at the PLC high-limit temperature $T_{\text{oil}} = 75^\circ\text{C}$ (Table 14) — the highest temperature T_{10} can sustain under passive solar-loop operation, and therefore the worst-case (highest) standing-loss condition the auxiliary/solar supply must be sized to overcome.

$$\Delta T_{\text{vessel}} = 75 - 18.1 = 56.9 \text{ K} \quad T_f = \frac{75 + 18.1}{2} = 46.55^\circ\text{C} \quad (28)$$

Air properties at $T_f = 46.55^\circ\text{C}$ (interpolated between the 25°C and 50°C property-table rows used in Section 4.3, $f = 0.862$): $k_{\text{air}} = 0.02710 \text{ W m}^{-1} \text{ K}^{-1}$, $\mu_{\text{air}} = 1.947 \times 10^{-5} \text{ Pa s}$, $\nu_{\text{air}} = 1.765 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, $\text{Pr}_{\text{air}} = 0.7237$, $\beta_{\text{air}} = 1/T_f(\text{K}) = 3.128 \times 10^{-3} \text{ K}^{-1}$.

Cylindrical wall.

$$Ra_L = \frac{9.81 \times 3.128 \times 10^{-3} \times 56.9 \times 2.20^3}{(1.765 \times 10^{-5})^2} \times 0.7237 = 4.317 \times 10^{10} \quad (29)$$

$$\overline{Nu}_L = \left[0.825 + \frac{0.387 \times (4.317 \times 10^{10})^{1/6}}{1.191} \right]^2 = 402.7 \Rightarrow h_{\text{ext,cyl}} = \frac{402.7 \times 0.02710}{2.20} = 4.960 \text{ W m}^{-2} \text{ K}^{-1} \quad (30)$$

Flat base ($L_c = 0.58 \text{ m}$).

$$Ra_{L_c} = \frac{9.81 \times 3.128 \times 10^{-3} \times 56.9 \times 0.58^3}{(1.765 \times 10^{-5})^2} \times 0.7237 = 7.910 \times 10^8 \quad (31)$$

$$\overline{Nu}_{L_c} = 0.27 \times (7.910 \times 10^8)^{0.25} = 45.28 \Rightarrow h_{\text{ext,base}} = \frac{45.28 \times 0.02710}{0.58} = 2.115 \text{ W m}^{-2} \text{ K}^{-1} \quad (32)$$

Overall U and loss. At 75°C the oil is fully liquid, so the established liquid-phase oil-side film coefficient applies: $h_{\text{in,vessel}} = h_o = 138.9 \text{ W m}^{-2} \text{ K}^{-1}$ (Section 7).

$$U_{\text{ves,cyl,75}} = \left(\frac{1}{138.9} + 1.20 \times 10^{-4} + \frac{1}{4.960} \right)^{-1} = 4.786 \text{ W m}^{-2} \text{ K}^{-1} \quad (33)$$

$$U_{\text{ves,base,75}} = \left(\frac{1}{138.9} + 1.20 \times 10^{-4} + \frac{1}{2.115} \right)^{-1} = 2.083 \text{ W m}^{-2} \text{ K}^{-1} \quad (34)$$

$$\dot{Q}_{\text{loss,cyl,75}} = 4.786 \times 16.035 \times 56.9 = 4366.7 \text{ W} \quad (35)$$

$$\dot{Q}_{\text{loss,base,75}} = 2.083 \times 4.229 \times 56.9 = 501.1 \text{ W} \quad (36)$$

$$\dot{Q}_{\text{loss,max}} = 4366.7 + 501.1 = 4867.8 \text{ W} \approx 4.87 \text{ kW} \quad (37)$$

This is the conservative, worst-case standing-loss figure carried forward into the Q_{target} definition of Section 8. It applies regardless of which solar-driver scenario (75 °C or 70 °C) is being evaluated, since the PLC high-limit of 75 °C is a fixed property of the tank/control system, not of the instantaneous driver temperature.

5 Coil Configuration

5.1 Material and Dimensions

A single supply pipe size is used throughout the loop: **1-inch NPS Schedule 40 CPVC** ($d_{i,\text{sup}} = 26.64 \text{ mm}$). The *coil* (water-side, submerged) pipe is treated as the principal design variable and is evaluated across four standard bores in SS 304 Schedule 10S, all sharing the same 2.77 mm wall thickness (consistent with the established Schedule 10S dimensions for this NPS range):

Table 3: Coil pipe dimensions and thermal conductivity (SS 304 Sch 10S).

Parameter	1-inch	1.25-inch	1.5-inch	2-inch
Outer diameter, d_o	33.40 mm	42.16 mm	48.26 mm	60.33 mm
Inner diameter, d_i	27.86 mm	36.62 mm	42.72 mm	54.79 mm
Wall thickness	2.77 mm			
d_o/d_i	1.1989	1.1513	1.1297	1.1011

Thermal conductivity (SS 304): $k_{\text{SS}} = 16.3 \text{ W m}^{-1} \text{ K}^{-1}$.

5.2 Helix Geometry

Hot water is fed from the *bottom* of the helix (counter-flow to the rising natural convection currents of the oil), maximising the temperature driving force along the entire coil length. The tank geometry fixes the helix diameter; this is unchanged by coil bore.

Table 4: Coil helix geometry — maximum (envelope) values.

Parameter	T_{10} (10 kL)	T_5 (5 kL)
Helix diameter, D_{helix} (m)	2.072	1.711
Maximum practical coil length, L_{max} (m)	90.0	65.0

Note — *length is now a design variable, not a fixed input.* L_{max} above is the geometric *ceiling* set by the helix envelope that fits inside each vessel (unchanged from the original sizing study). The *actual* coil length required to meet the new heat-duty target, L_{required} , is solved for explicitly in Section 9 using the closed-form ε -NTU inversion, and is in every case shorter than this ceiling. Coil surface area is therefore no longer quoted as a single fixed number — it follows from whichever $(d_o, L_{\text{required}})$ pair is ultimately selected.

5.3 Wall Conduction Resistance of Coil

$$R_{\text{wall}} = \frac{d_o \ln(d_o/d_i)}{2 k_{\text{SS}}} \quad (38)$$

Table 5: Coil wall conduction resistance by bore.

	1-inch	1.25-inch	1.5-inch	2-inch
$R_{\text{wall}} (\times 10^{-4} \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1})$	1.857	1.822	1.805	1.782

6 Water-Side Hydraulics and Convection

With a single 1-inch CPVC supply pipe, the system mass flow rate and water capacity rate are fixed quantities, independent of which coil bore is ultimately selected — only the in-coil velocity (and hence h_i) changes with bore, since mass is conserved through the loop.

Water properties at $T_{\text{hot}} = 75^\circ\text{C}$: $\rho_w = 974.9 \text{ kg m}^{-3}$, $C_{p,w} = 4193 \text{ J kg}^{-1} \text{ K}^{-1}$, $\mu_w = 3.78 \times 10^{-4} \text{ Pa s}$, $k_w = 0.6636 \text{ W m}^{-1} \text{ K}^{-1}$, $\text{Pr}_w = 2.39$.

6.1 Flow Rate and Water Capacity Rate

At the design supply velocity $V_{\text{max}} = 1.5 \text{ m s}^{-1}$:

$$\dot{m}_w = \rho_w \frac{\pi d_{i,\text{sup}}^2}{4} V_{\text{max}} = 974.9 \times \frac{\pi \times 0.02664^2}{4} \times 1.5 = 0.815 \text{ kg s}^{-1} \quad (39)$$

$$C_w = \dot{m}_w C_{p,w} = 0.815 \times 4193 = 3417.7 \text{ W K}^{-1} \quad (40)$$

This single value of C_w (and hence C_{min} , since $C_r = 0$ for the T_{10} melting duty) applies to every coil-bore option evaluated below.

6.2 Coil-Side Velocity and Reynolds Number

$$Re_{\text{coil}} = \frac{4 \dot{m}_w}{\pi d_i \mu_w} \quad (41)$$

Table 6: Coil-side flow conditions, single 1-inch supply, by coil bore.

Coil bore	1-inch	1.25-inch	1.5-inch	2-inch
Re_{coil}	98,548	74,974	64,268	50,110
Regime	turbulent	turbulent	turbulent	turbulent

6.3 Inner Heat Transfer Coefficient

Gnielinski correlation with Kubair–Kuloor helical enhancement, evaluated at each coil bore's Re_{coil} , $D_{\text{helix}, T_{10}} = 2.072 \text{ m}$:

$$f = (0.79 \ln Re_{\text{coil}} - 1.64)^{-2} \quad (42)$$

$$Nu_{\text{str}} = \frac{(f/8)(Re_{\text{coil}} - 1000) Pr_w}{1 + 12.7 \sqrt{f/8} (Pr_w^{2/3} - 1)} \quad (43)$$

$$Nu_{\text{helix}} = Nu_{\text{str}} \left(1 + 3.54 \frac{d_i}{D_{\text{helix}}} \right) \quad (44)$$

$$h_{i,w} = \frac{Nu_{\text{helix}} k_w}{d_i} \quad (45)$$

Table 7: Inner heat transfer coefficient by coil bore — T₁₀ helix.

Coil bore	1-inch	1.25-inch	1.5-inch	2-inch
f	0.01805	0.01914	0.01981	0.02095
Nu_{str}	356.6	284.1	249.9	203.3
Nu_{helix}	373.5	301.8	268.2	222.3
$h_{i,w}$ (W/m ² .K)	8897	5470	4166	2693

As in the original analysis, the inner coefficient varies by more than $3\times$ across the bore range (driven purely by velocity, since mass flow is fixed), but Section 7 shows that this has only a marginal effect on the overall U — the outer oil-side resistance continues to dominate.

7 Two-Phase U -Value Determination

7.1 Fouling Allowances

Table 8: Fouling resistances applied per TEMA standards.

Surface	Condition	R_f ($\text{m}^2 \cdot \text{K} \cdot \text{W}^{-1}$)
Inner (water side)	Soft water, low cycling	1.0×10^{-4}
Outer (palm oil side)	Palm oil at 60 °C, convection	9.0×10^{-4}

7.2 Overall Heat Transfer Coefficient

The general resistance equation (referred to the outer surface area) is:

$$\frac{1}{U} = \frac{1}{h_{i,w}} \cdot \frac{d_o}{d_i} + R_{f,i} \cdot \frac{d_o}{d_i} + R_{\text{wall}} + R_{f,o} + \frac{1}{h_o} \quad (46)$$

where $h_o = 35 \text{ W m}^{-2} \text{ K}^{-1}$ (solid/melting phase) and $h_o = 138.9 \text{ W m}^{-2} \text{ K}^{-1}$ (liquid phase) — both independent of pipe size, since they are governed by tank-side natural-convection geometry.

Table 9: Overall U by coil bore, single 1-inch supply pipe.

Coil bore	1-inch	1.25-inch	1.5-inch	2-inch
$1/(h_i \cdot d_o/d_i)$ ($\times 10^{-3}$)	0.135	0.211	0.271	0.408
$R_{f,i} \cdot (d_o/d_i)$ ($\times 10^{-3}$)	0.120	0.115	0.113	0.110
R_{wall} ($\times 10^{-3}$)	0.186	0.182	0.181	0.178
$R_{f,o}$ ($\times 10^{-3}$)	0.900			
U_{solid} ($\text{W/m}^2 \cdot \text{K}$)	33.43	33.36	33.29	33.15
U_{liquid} ($\text{W/m}^2 \cdot \text{K}$)	117.10	116.18	115.42	113.68

The spread in U_{solid} across the entire 1 inch–2 inch bore range is only $0.28 \text{ W m}^{-2} \text{ K}^{-1}$ (0.8%), and in U_{liquid} only $3.4 \text{ W m}^{-2} \text{ K}^{-1}$ (2.9%). The outer oil-side resistance ($1/h_o$) accounts for >95% of $1/U_{\text{solid}}$ in every case, confirming — as in the original study — that coil bore has almost no effect on the overall heat transfer coefficient. **The coil-sizing decision is therefore governed almost entirely by surface area (i.e. by length and bore), not by U .** This is precisely why coil length can now be treated as a free design variable and solved for directly.

8 Coil-Sizing Target: Q_{batch} and Q_{target}

8.1 Reframing the Design Problem

The original sizing approach fixed the coil at its maximum practical length ($L_{\text{max}} = 90 \text{ m}$ for T_{10}) and then asked: *how much oil can this coil melt, and how quickly?* This proposal inverts the logic: a heat-duty *target*, Q_{target} , is defined directly from the production

mandate and the worst-case standing loss, and the **minimum coil length required to deliver that target** is then solved for in closed form. This yields a shorter, more economical coil wherever the target is comfortably below what the maximum-length coil could deliver — converting unused thermal margin directly into a pipe-length (and cost) saving.

8.2 Q_{batch} — Heat Duty for the 2 t/day Mandate

Consistent with the "single-tank capacity" basis used throughout this proposal (i.e. the conservative assumption that one T_{10} tank alone must be able to cover the full daily mandate), Q_{batch} is defined as the *average* heat duty required to melt 2t of palm oil within the full 24 hr daily allowance:

$$Q_{\text{batch}} = \frac{E_{2t}}{t_{\text{day}}} = \frac{2000 \text{ kg} \times q_{\text{specific}}}{24 \times 3600 \text{ s}} = \frac{2000 \times 185,800}{86,400} = 4301 \text{ W} \approx 4.30 \text{ kW} \quad (47)$$

where $q_{\text{specific}} = 185,800 \text{ J kg}^{-1}$ is the melting specific energy established in Section 10 ($C_{p,s}\Delta T + L_f$, $\Delta T = 18.1 \rightarrow 36^\circ\text{C}$). This is a deliberately conservative basis: it is the *minimum* sustained power a single tank's coil must deliver, net of losses, to just meet the daily quota with zero spare time — i.e. the floor on acceptable performance, not a target throughput.

8.3 $Q_{\text{loss,max}}$ — Worst-Case Standing Loss

From Section 4.6, the heat loss from an uninsulated T_{10} tank at its PLC high-limit temperature of 75°C is:

$$Q_{\text{loss,max}} \approx 4.87 \text{ kW} \quad (48)$$

8.4 Q_{target}

$$Q_{\text{target}} = Q_{\text{batch}} + Q_{\text{loss,max}} = 4.30 + 4.87 = 9.17 \text{ kW} \quad (49)$$

Interpretation: a coil sized to deliver $Q_{\text{target}} = 9.17 \text{ kW}$ of *gross* heat transfer at melt-phase conditions will, after the tank sheds heat at its own worst-case rate, still net out $Q_{\text{batch}} = 4.30 \text{ kW}$ into the oil — exactly enough to clear the 2t daily mandate in 24 hr, even if the tank happens to be sitting at its 75°C high-limit (the most thermally leaky condition it can be in) at the time.

9 Closed-Form Coil Length Required to Meet Q_{target}

9.1 Derivation

For the T_{10} melting duty, $C_r = 0$ (the oil absorbs heat isothermally at $T_{\text{melt}} = 36^\circ\text{C}$), so $\varepsilon = 1 - e^{-NTU}$ and $C_{\text{min}} = C_w$. The transmissible heat is:

$$Q = \varepsilon C_{\text{min}} \Delta T = (1 - e^{-UA/C_{\text{min}}}) C_{\text{min}} \Delta T \quad (50)$$

Setting $Q = Q_{\text{target}}$ and solving for the required area A :

$$e^{-UA/C_{\min}} = 1 - \frac{Q_{\text{target}}}{C_{\min}\Delta T} = \frac{C_{\min}\Delta T - Q_{\text{target}}}{C_{\min}\Delta T} \quad (51)$$

$$A_{\text{required}} = \ln\left(\frac{C_{\min}\Delta T}{C_{\min}\Delta T - Q_{\text{target}}}\right) \cdot \frac{C_{\min}}{U} \quad (52)$$

Substituting the geometric identity $A = \pi d_o L$:

$$L_{\text{required}} = \ln\left(\frac{C_{\min}\Delta T}{C_{\min}\Delta T - Q_{\text{target}}}\right) \cdot \frac{C_{\min}}{U \cdot \pi d_o} \quad (53)$$

where $\Delta T = T_{\text{hot}} - T_{\text{melt}}$, $U = U_{\text{solid}}$ (the melting-phase value — the system bottleneck, per the original conclusions), and $C_{\min} = C_w = 3417.7 \text{ W K}^{-1}$ (Section 6.1, single 1-inch supply, both driver scenarios).

9.2 Required Effectiveness and NTU

Table 10: Required effectiveness and NTU to meet $Q_{\text{target}} = 9.17 \text{ kW}$ (coil-bore-independent).

Driver scenario	$T_{\text{hot}} = 75^\circ\text{C}$	$T_{\text{hot}} = 70^\circ\text{C}$
$\Delta T = T_{\text{hot}} - T_{\text{melt}}$ (K)	39	34
$C_{\min}\Delta T$ (kW)	133.3	116.2
$\varepsilon_{\text{required}}$	0.0688	0.0789
NTU_{required}	0.0713	0.0822

Both required effectiveness values are small (under 8%), confirming that the original 90 m coil — sized for an effectiveness of 5–12% delivering 10–16 kW — was substantially oversized relative to the true daily-quota requirement.

9.3 Required Coil Length by Bore

Table 11: Required coil length L_{required} to deliver $Q_{\text{target}} = 9.17 \text{ kW}$, vs. the 90 m geometric ceiling, T_{10} .

Coil bore	1-inch	1.25-inch	1.5-inch	2-inch
U_{solid} (W/m ² .K)	33.43	33.36	33.29	33.15
L_{required} at 75°C (m)	69.4	55.1	48.3	38.8
Turns required, $N = L/(\pi D_{\text{helix}})$	10.7	8.5	7.4	6.0
% of 90 m ceiling used	77%	61%	54%	43%
L_{required} at 70°C (m)	80.1	63.6	55.6	44.7
Turns required, $N = L/(\pi D_{\text{helix}})$	12.3	9.8	8.5	6.9
% of 90 m ceiling used	89%	71%	62%	50%

All four bore options satisfy Q_{target} within the 90 m geometric ceiling, in both driver scenarios. The 1-inch bore (matching the supply pipe — simplest procurement, one pipe spec end-to-end) requires 69.4 m at 75 °C drive and 80.1 m at the 70 °C worst case, leaving only ≈ 10 m (11%) of headroom in the worst case. The larger bores trade a (negligibly) lower U for a much larger area per metre, cutting required length by up to 44% (2-inch vs. 1-inch at 75 °C). **Recommendation:** specify the **1.5-inch coil at $L \approx 56$ m** (rounded up from 55.6 m for installation tolerance) as the design point — it keeps utilisation at a comfortable 62% of the geometric ceiling even in the worst-case 70 °C scenario, while still cutting the original 90 m coil specification by $\approx 38\%$, a direct pipe-length (and cost) saving. The 1-inch option remains available as a lower-cost, single-pipe-spec alternative if the $\approx 11\%$ worst-case margin is judged acceptable.

Important caveat. L_{required} is solved to deliver *exactly* Q_{target} at the design driver temperature — it contains no additional margin beyond the conservative choices already built into Q_{target} itself (worst-case standing loss at the 75 °C high-limit, and zero slack on the 24-hour daily quota). Any coil shorter than L_{required} will fail to clear 2 t/day once the tank reaches its high-limit temperature. The percentages above should be read as the margin *within* the geometric/installation ceiling, not as overall safety margin on production — for the latter, see Section 10.2.

10 Energy Budget and Melt Time

10.1 Total Energy Requirement — Full 10 kL Tank

$$E_{\text{sensible}} = 318.6 \text{ MJ} \quad E_{\text{latent}} = 1335.0 \text{ MJ} \quad E_{\text{total}} = 1653.6 \text{ MJ} \quad (54)$$

$$\text{Specific energy: } q_{\text{specific}} = C_{p,s} \Delta T + L_f = 2000 \times 17.9 + 150,000 = 185.800 \text{ J kg}^{-1}.$$

10.2 Melt Time at the Required Coil Length

Because each coil in Section 9 is sized to deliver exactly Q_{target} (gross) — i.e. exactly $Q_{\text{batch}} = 4.30 \text{ kW}$ net of the worst-case standing loss — the corresponding melt time for the full 1653.6 MJ batch, and for the 2 t daily mandate specifically, is **identical across all four bore options by construction**:

$$t_{\text{melt},2t} = \frac{E_{2t}}{Q_{\text{batch}}} = \frac{371.6 \times 10^6}{4301} = 86.400 \text{ s} = 24.0 \text{ hr} \quad (\text{exactly the daily allowance, by design}) \quad (55)$$

$$t_{\text{melt,full}} = \frac{E_{\text{total}}}{Q_{\text{batch}}} = \frac{1653.6 \times 10^6}{4301} = 384.470 \text{ s} \approx 106.8 \text{ hr} \quad (56)$$

This is the expected consequence of sizing to the floor of the daily mandate with zero slack (Section 8): a *single* T_{10} tank, run continuously at its Q_{target} -rated coil, clears its 2 t share in exactly 24 hr when standing losses are at their worst-case (high-limit) value. In practice, losses are below the 75 °C high-limit value for most of the melt cycle (loss is lower while the tank is still cool, per Section 4), so actual melt time will be somewhat faster than this conservative figure. With **three** T_{10} tanks staggered, the system-level daily capacity is $3 \times 2 \text{ t} = 6 \text{ t/day}$ at this conservative, zero-margin design point — a 3× margin over the 2 t/day mandate purely from running all three tanks, even before accounting for the faster-than-worst-case loss behaviour noted above.

10.3 Solar Circuit Characterization — Kathmandu Conditions

Average operational irradiance: $G = 750 \text{ W m}^{-2}$, peak sun hours: $PSH = 4.61 \text{ hr}$, ambient temperature: $T_a = 18.1 \text{ °C}$, $\Delta T = T_f - T_a = 56.9 \text{ °C}$.

Flat Plate Collector (FPC)

$N = 15$ panels, $A_{c,\text{FPC}} = 22.674 \text{ m}^2$, $\eta_0 = 0.75$, $a_1 = 3.50 \text{ W m}^{-2} \text{ K}^{-1}$, $a_2 = 0.015 \text{ W m}^{-2} \text{ K}^{-2}$.

$$\eta_{\text{FPC}} = 0.75 - \frac{3.50 \times 56.9}{750} - \frac{0.015 \times 56.9^2}{750} = 0.4197 = 41.97 \% \quad (57)$$

$$\dot{Q}_{\text{peak,FPC}} = 7.14 \text{ kW} \quad \mathbf{E}_{\text{solar,FPC}} = 118.45 \text{ MJ} \quad (58)$$

Evacuated Tube Collector (ETC)

$N = 5$ units, $A_{c,\text{ETC}} = 32.50 \text{ m}^2$, $\eta_0 = 0.70$, $a_1 = 1.50 \text{ W m}^{-2} \text{ K}^{-1}$, $a_2 = 0.005 \text{ W m}^{-2} \text{ K}^{-2}$.

$$\eta_{\text{ETC}} = 0.70 - \frac{1.50 \times 56.9}{750} - \frac{0.005 \times 56.9^2}{750} = 0.5646 = 56.46 \% \quad (59)$$

$$\dot{Q}_{\text{peak,ETC}} = 13.76 \text{ kW} \quad \mathbf{E}_{\text{solar,ETC}} = 228.40 \text{ MJ} \quad (60)$$

10.4 Solar Integration — Full Batch and 2-Tonne Target

$$E_{\text{total,full}} = 1653.6 \text{ MJ} \quad (61)$$

$$E_{\text{total,2t}} = 2000 \times 185,800 \times 10^{-6} = 371.6 \text{ MJ} \quad (62)$$

Config.	E_{solar} (MJ)	Batch coverage (%)		$E_{\text{auxiliary}}$ (MJ)	
		Full batch	2-tonne	Full batch	2-tonne
FPC	118.45	7.16	31.88	1535.15	253.15
ETC	228.40	13.81	61.46	1425.20	143.20

The ETC configuration covers 61.46% of the 2 t/day minimum production energy demand from solar alone, reducing auxiliary consumption to 143.20 MJ — a 43.4% reduction versus the FPC auxiliary requirement. These figures are independent of coil bore and supply-pipe choice.

11 T₅ Replenishment Tank — Sensible Heating

11.1 Process Conditions

Palm oil arrives from T₁₀ via HPP at $\approx 36^\circ\text{C}$ and is trim-heated to the process target of 60°C . The oil-side capacity rate is:

$$C_{\text{po}} = \frac{2000}{20 \times 3600} \times 2100 = 58.33 \text{ W K}^{-1} \quad (63)$$

Since $C_{\text{po}} \ll C_w$ for every coil bore evaluated, $C_r \approx 0$ and $C_{\text{min}} = C_{\text{po}}$ regardless of bore — the same simplification used in the original analysis, now also independent of supply-pipe choice since only one supply pipe remains in the design.

11.2 Minimum Coil Length to Reach the 60°C Process Target

Whereas T₁₀'s coil is sized to deliver a partial-effectiveness duty (Q_{target}), T₅'s minimum requirement is different in kind: the oil must reach *at least* the 60°C process target, i.e. $Q_{\text{design,T5}} = C_{\text{po}} \times (60 - 36) = 1400 \text{ W}$ exactly. The same closed-form inversion of Section 9 applies, with $C_{\text{min}} = C_{\text{po}}$, $U = U_{\text{liquid}}$, and $D_{\text{helix,T5}} = 1.711 \text{ m}$:

Table 12: T₅ — minimum coil length to exactly reach 60°C , by coil bore.

Coil bore	1-inch	1.25-inch	1.5-inch	2-inch
U_{liquid} (W/m ² .K)	117.12	116.22	115.47	113.77
$\varepsilon_{\text{required}}$ (both drivers)	0.615 at 75°C ; 0.706 at 70°C			
L_{required} at 75°C (m)	4.5	3.6	3.2	2.6
L_{required} at 70°C (m)	5.8	4.6	4.1	3.3

T₅ is dramatically over-specified relative to its actual duty. Even the smallest bore needs only ≈ 6 m of coil (against the 65 m geometric ceiling — under 9% utilisation) to guarantee the 60 °C process target is reached. This confirms and quantifies the original finding that T₅ is not capacity-limiting: at the full 65 m length, the oil outlet temperature saturates at the hot-water supply temperature (≈ 75 °C, i.e., 15 °C *above* the process target), which is precisely why a duty-cycle loop or mixing valve was identified as necessary in the original control logic. A coil specified close to the L_{required} values above — rather than the full 65 m ceiling — would land much closer to the 60 °C target under continuous flow, reducing (though not eliminating, given start-up and turn-down transients) reliance on that control loop. No further optimisation of T₅ is pursued here, as it is not capacity-limiting for the 2 t/day mandate.

12 Process Control Architecture

12.1 Overview

The control system operates on a **demand-driven** logic: T₅ is the process master. A batch cycle in T₁₀ initiates *only* when T₅ level or temperature drops below its setpoint. This is fundamentally different from timed batch control.

12.2 Sensor Naming Convention

Table 13: Sensor and signal naming convention used throughout this section.

Symbol	Meaning
θ	All temperature sensors (e.g. θ_S — T ₅ temperature, θ_{Sol} — solar loop temperature, θ_H — thermal battery temperature, θ_{10} — T ₁₀ tank temperature)
α	All float / level sensors (e.g. α_S — T ₅ level, α — T ₁₀ tank level)
$\delta_X, \delta_Y, \delta_Z$	Distribution demand signals from downstream HMIs X, Y, Z

12.3 Derivation of the Dynamic Upper Thermal Boundary (T_f)

To ensure process safety and maintain structural integrity, the system utilizes a dynamic upper thermal boundary constraint ($T_f \leq 55$ °C). While the lower limit (40 °C) is explicitly measured at the tank, the upper threshold fluctuates depending on the thermodynamic capacities of the mixing mediums: water (w), stainless steel structural components (ss), and the process liquid palm oil (p).

The final equilibrium temperature, T_f , is governed by the conservation of energy principle, assuming complete (100%) heat transfer under worst-case thermal absorption conditions:

$$Q_{\text{lost}} = Q_{\text{gained}} \quad (64)$$

The total heat lost by the water and stainless steel component mass cooling from an initial temperature x to T_f is expressed as:

$$Q_{\text{lost}} = (m_w c_w + m_{ss} c_{ss})(x - T_f) \quad (65)$$

Where:

- m_w = mass of water ≈ 90 kg (derived from 90 L)
- c_w = specific heat capacity of water ≈ 4184 J/kg $\cdot$ $^{\circ}$ C
- m_{ss} = mass of stainless steel structure = 205 kg
- c_{ss} = specific heat capacity of stainless steel ≈ 500 J/kg $\cdot$ $^{\circ}$ C

Substituting these constant nominal values yields:

$$\begin{aligned} Q_{\text{lost}} &= [(90 \times 4184) + (205 \times 500)] (x - T_f) \\ Q_{\text{lost}} &= (376560 + 102500)(x - T_f) \\ Q_{\text{lost}} &= 479060(x - T_f) \end{aligned} \tag{66}$$

Concurrently, the heat energy gained by the palm oil mass (m_p) shifting from an initial temperature y up to equilibrium T_f is calculated using the standard specific heat capacity of palm oil ($c_p \approx 2000$ J/kg $\cdot$ $^{\circ}$ C):

$$Q_{\text{gained}} = m_p c_p (T_f - y) = 2000 m_p (T_f - y) \tag{67}$$

Equating energy loss to energy gain allows the isolation of the equilibrium parameter:

$$479060(x - T_f) = 2000 m_p (T_f - y) \tag{68}$$

Expanding and solving for the final system equilibrium temperature (T_f) yields the dynamic tracking function:

$$T_f(x, y, m_p) = \frac{479060x + 2000m_p y}{479060 + 2000m_p} \tag{69}$$

Dividing by the common scalar factor of 2000 simplifies the expression for efficient implementation within runtime PLC memory registers:

$$T_f(x, y, m_p) = \frac{239.53x + m_p y}{239.53 + m_p} \tag{70}$$

The active safety logic system monitors continuously evaluated inputs (x, y, m_p) and triggers an interlock cycle whenever the function yields $T_f(x, y, m_p) > 55^{\circ}\text{C}$.

12.4 Control Loops and Setpoints

Table 14: PLC control loops, sensors, and actions with dynamic thermal boundaries.

Condition	Priority / Mode	Action
$\theta_S < 40^\circ\text{C}$	Priority 1	Heat T_5 (open coil loop valve to T_5)
$40^\circ\text{C} \leq \theta_S \leq T_f(x, y, m_p)$	—	Normal operating band; no action required
$T_f(x, y, m_p) > 55^\circ\text{C}$	Safety / Upper Limit	Dynamic cut-off triggered; throttle process or divert flow
$\alpha_S < 50\%$	Priority 2 (after temp)	Trigger HPP to transfer liquid palm oil, $T_{10} \rightarrow T_5$
$\theta_{\text{Sol}} < 70^\circ\text{C}$ and $\theta_H > 70^\circ\text{C}$	—	Bypass solar loop
$\theta_{\text{Sol}} < 70^\circ\text{C}$ and $\theta_H < 70^\circ\text{C}$	—	Bypass solar loop and activate immersion elements
$\theta_H < 60^\circ\text{C}$ with elements on	Alarm	Alarm; circulate loop until $\theta_H < 36^\circ\text{C}$, then enter thermal recharge
Thermal recharge mode	Alarm	Alarm; cease flow; apply full auxiliary heat output; resume heating the highest-ranked T_{10} tank when $\theta_{\text{Sol}} > 70^\circ\text{C}$
T_{10} surplus (T_5 not in demand)	—	Route hot water to T_{10} until $\alpha_S > 90\%$
T_{10} high-limit	Safety	Close coil valve; alarm if $T_{10} > 75^\circ\text{C}$

Note on thermal recharge: Recharge mode is entered only after the $\theta_H < 60^\circ\text{C}$ alarm condition persists and the battery has been allowed to circulate down to $\theta_H < 36^\circ\text{C}$. During recharge, flow to all downstream loops ceases and the full auxiliary heating capacity is dedicated to restoring θ_H . If solar availability returns during recharge ($\theta_{\text{Sol}} > 70^\circ\text{C}$), the PLC may redirect heating priority to the highest-ranked T_{10} tank to make best use of the available thermal input.

12.5 Tank Selection Scoring System

Because three T_{10} tanks (A, B, C) compete for both water-heating priority and oil draw-off priority, each tank is continuously assigned two scores by the PLC:

$$S_{\text{casual}} = 0.6 \alpha + 0.4 \theta \quad (\text{volume-weighted; used for normal draw}) \quad (71)$$

$$S_{\text{immediate}} = 0.4 \alpha + 0.6 \theta \quad (\text{temperature-weighted; used for urgent draw}) \quad (72)$$

where α and θ are the normalised level and temperature readings of the individual T_{10} tank in question.

- Under normal operating conditions, the PLC ranks $T_{10,A}$, $T_{10,B}$, and $T_{10,C}$ by S_{casual} ; the highest-scored tank is drawn from first.

- Under urgent/low-buffer conditions, ranking switches to $S_{\text{immediate}}$, weighting temperature more heavily so that the nearest-to-target tank is prioritised.
- Water-heating priority follows the *same* ranking: the highest-scored tank (by the applicable score) also receives the next allocation of hot water from the thermal battery.

12.6 Oil Distribution Logic

T_5 supplies liquid palm oil to three downstream process lines (X, Y, Z) through three proportional control valves. Each line reports a continuous demand signal (δ_X , δ_Y , δ_Z) from its respective HMI.

Table 15: T_5 outlet distribution logic.

Condition	Action
$\delta_X + \delta_Y + \delta_Z \leq \text{Supply}$	All three valves open proportionally; free flow to all lines
$\delta_X + \delta_Y + \delta_Z > \text{Supply}$	Priority queue applies: line X is served to its full demand first, then Y, then Z, until supply is exhausted

- The $X \rightarrow Y \rightarrow Z$ priority order is configurable by the PLC programmer via the HMI; it is **not** hardwired and may be re-ordered to match changing production priorities.
- All three outlet valves are proportional (modulating) control valves, not simple on/off devices, allowing partial opening to match any demand level below full flow.

12.7 Priority Logic and Solar Circulation Modes

The PLC operates on the following priority hierarchy:

1. **T_5 temperature** (θ_S) is the primary priority. If $\theta_S < 40^\circ\text{C}$, T_5 's coil loop is active.
2. When T_5 is satisfied, the water loop is directed to whichever T_{10} tank ranks highest by the applicable scoring system (Section 12.5).
3. **Three solar circulation modes govern loop routing:**
 - *Through-solar*: Loop water passes through the solar collector array, drawing active heat from the collectors into the system. This mode is used when the collectors are producing useful heat ($\theta_{\text{Sol}} \geq 70^\circ\text{C}$) and the reservoir is below target.
 - *Bypass-solar*: The solar collector array is bypassed entirely. This mode is engaged when $\theta_{\text{Sol}} < 70^\circ\text{C}$ (collectors not generating useful heat) or when the reservoir is already at or above T_{hot} .
 - *Quick-solar cycle*: A short gravity-assisted circulation stroke used to prime the solar loop pump at start-up, confirming flow before committing to through-solar operation.
4. **Thermal battery interlock**: If $\theta_H < 60^\circ\text{C}$ with immersion elements already energised, the alarm and recharge sequence of Section 12.3 governs.

12.8 Batch Release Condition

A T_{10} batch is *only released* (HPP activated to T_5) when:

$$\alpha_S < 50\% \quad \text{AND} \quad \theta_{10}(\text{highest-ranked tank}) \geq 36^\circ\text{C} \quad (73)$$

The tank selected to supply the batch is the one ranked highest by $S_{\text{immediate}}$ at the moment of release. This demand-pull logic prevents premature extraction of unmelted oil and ensures T_5 never starves the process lines.

13 Design Validation Summary

Table 16: Design parameter summary — vessel, coil, and thermal performance, single 1-inch supply pipe, by coil bore.

Parameter	1-inch	1.25-inch	1.5-inch	2-inch
<i>Vessel — T_{10} (common to all cases)</i>				
External surface area, A_{vessel} (m ²)		20.264		
Tank heat loss, melt phase at 36°C (W)		1071.6		
Tank heat loss, high-limit at 75°C (W)		4867.8		
<i>Coil — T_{10} (1-inch CPVC supply, single pipe size)</i>				
Coil material		SS 304 Sch 10S		
Coil NPS (inch)	1	1.25	1.5	2
Re_{coil}	98,548	74,974	64,268	50,110
Water-side h_i (W/m ² ·K)	8897	5470	4166	2693
U_{solid} (W/m ² ·K)	33.43	33.36	33.29	33.15
U_{liquid} (W/m ² ·K)	117.10	116.18	115.42	113.68
<i>Coil-Sizing Target (Section 8, common to all cases)</i>				
Q_{batch} (kW)		4.30		
$Q_{\text{loss,max}}$ (kW)		4.87		
Q_{target} (kW)		9.17		
<i>Required Coil Length (Section 9)</i>				
L_{required} at 75°C (m)	69.4	55.1	48.3	38.8
L_{required} at 70°C (m)	80.1	63.6	55.6	44.7
Geometric ceiling, L_{max} (m)		90.0		
% of ceiling used (worst case, 70°C)	89%	71%	62%	50%
<i>Energy Budget — Full 10 kL Batch (common to all cases)</i>				
E_{sensible} , 18.1→36°C (MJ)		318.6		
E_{latent} , fusion (MJ)		1335.0		
E_{total} (MJ)		1653.6		
$t_{\text{melt,2t}}$ at Q_{batch} (hr)		24.0 (by design)		

Table 17: Design parameter summary — production capacity, solar integration, and T_5 replenishment tank.

Parameter	1-inch	1.25-inch	1.5-inch	2-inch
<i>System-Level Daily Capacity (3 tanks staggered, zero-margin design point)</i>				
Single-tank daily capacity, Q_{batch} basis (t/day)	2.0 (by design, all bores)			
3-tank staggered capacity (t/day)	6.0			
Margin vs. 2 t/day mandate	3.0×			
<i>Solar Integration (common to all cases)</i>				
FPC solar yield, E_{solar} (MJ)	118.45			
ETC solar yield, E_{solar} (MJ)	228.40			
FPC coverage — 2-tonne target (%)	31.88			
ETC coverage — 2-tonne target (%)	61.46			
Auxiliary energy, ETC, 2-t target (MJ)	143.20			
<i>T_5 Replenishment Tank (1-inch supply; geometric ceiling 65 m)</i>				
U_{liquid} (W/m ² ·K)	117.12	116.22	115.47	113.77
L_{required} to reach 60 °C at 70 °C drive (m)	5.8	4.6	4.1	3.3
% of 65 m ceiling used	9%	7%	6%	5%

14 Conclusions and Recommendations

1. **The coil-sizing problem has been inverted.** Rather than fixing the coil at its 90 m geometric ceiling and asking what throughput results, this revision defines a heat-duty target, $Q_{\text{target}} = Q_{\text{batch}} + Q_{\text{loss,max}} = 9.17 \text{ kW}$, directly from the 2 t/day mandate and the worst-case (high-limit, 75°C) standing loss, then solves in closed form for the minimum coil length that delivers it. Every bore evaluated (1–2 inch) requires substantially less than the 90 m ceiling — as little as 38.8 m (2-inch bore, 75°C drive) to 80.1 m (1-inch bore, 70°C worst case).
2. **Recommended design point: 1.5-inch coil at $L \approx 56 \text{ m}$.** This cuts the original 90 m specification by $\approx 38\%$ while retaining a comfortable $\approx 38\%$ margin against the geometric ceiling even in the worst-case 70°C driver scenario. The 1-inch bore is a viable lower-cost alternative (single pipe spec throughout the loop) if the tighter $\approx 11\%$ ceiling margin is acceptable.
3. **Supply-side simplification.** Standardising on a single 1 inch Schedule 40 CPVC supply pipe throughout the loop (removing the previous 1.5 inch/1 inch supply distinction) removes one entire axis of design complexity without affecting the U -value or NTU performance materially — Section 7 confirms the outer oil-side resistance dominates regardless of pipe size.
4. **The 2-tonne target is met with margin even at the conservative, zero-slack design point.** A single tank sized to Q_{target} clears its 2 t share in exactly 24 hr *at the worst-case standing loss*; with three tanks staggered the system delivers 6 t/day against the 2 t/day mandate — a $3\times$ margin — and actual performance will exceed this whenever the tank is below its 75°C high-limit, which is most of the melt cycle.
5. **T_5 remains dramatically over-specified.** Even at the smallest bore, only $\approx 6 \text{ m}$ of coil (9% of the 65 m ceiling) is needed to guarantee the 60°C process target is reached, confirming the original finding and quantifying the scale of the existing over-design. A mixing valve or duty-cycle loop remains advisable for transient/start-up conditions even at a right-sized coil length.
6. **Tank insulation remains recommended.** Insulating T_{10} would reduce both the melt-phase loss ($\approx 1.07 \text{ kW}$) and, more importantly for this revision, the high-limit standing loss that drives Q_{target} ($\approx 4.87 \text{ kW}$) — directly shrinking Q_{target} and therefore the required coil length for any bore chosen above.
7. **Thermal recharge interlock is critical.** The $\theta_H < 60^\circ\text{C}$ alarm and recharge sequence must be hardwired as a fail-safe, not software-only, to protect the pump from running against a hot but thermally depleted loop.
8. **ETC is the preferred collector technology; a break-even analysis is recommended before final procurement.** On thermal performance alone, the evacuated tube collector (ETC) configuration delivers **61.46%** solar coverage of the 2 t/day production energy demand versus **31.88%** for the flat plate collector (FPC) — a **29.6 percentage point** improvement, or approximately $1.93\times$ the solar yield for the same collector aperture area. Correspondingly, auxiliary energy consumption is reduced by 109.95 MJ per day under ETC operation. However, thermal performance figures alone do not constitute a procurement decision: ETC units carry a higher capital cost per

unit aperture area, and a formal break-even and lifecycle cost analysis — incorporating procurement and installation costs, auxiliary energy tariffs, degradation rates, and maintenance schedules — is recommended prior to final specification.

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